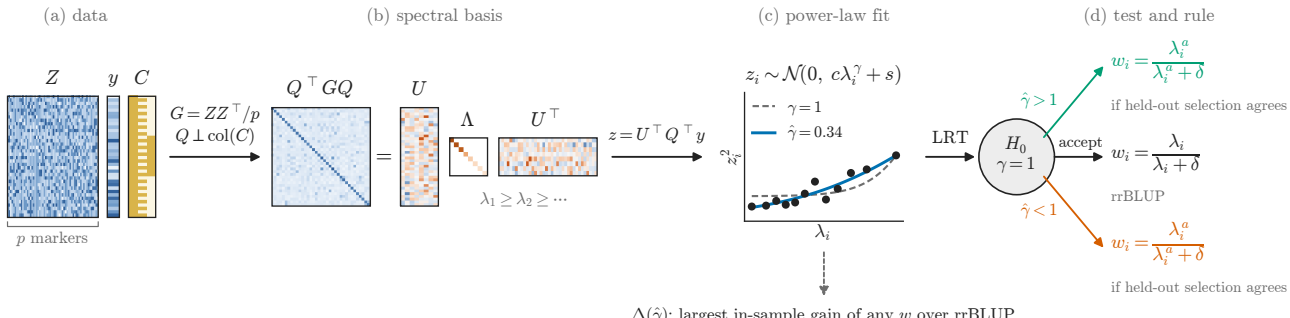
(a) data
(b) spectral basis
(c) power-law fit
(d) test and rule
Z
y
C
n
p markers
G = ZZ⊤/p
Q ⊥ col(C)
Q⊤GQ
=
U
Λ
U⊤
λ1 ≥ λ2 ≥ ···
z = U⊤Q⊤y
zi ∼ N(0, cλi^γ + s)
γ = 1
γ̂ = 0.34
zi²
λi
LRT
H0
γ = 1
accept
γ̂ > 1
γ̂ < 1
wi = λi^a / (λi^a + δ)
if held-out selection agrees
wi = λi / (λi + δ)
rrBLUP
wi = λi^a / (λi^a + δ)
if held-out selection agrees
Δ(γ̂): largest in-sample gain of any w over rrBLUP